TraceSmith — 系统架构 · GPU Trace 端到端数据流

Application → bindings → capture / state / format → replay & export → GPU backends

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

APPLICATION · 用户入口

C++ / Python Application
嵌入业务进程

CLI
profile · record · view · export

External UI
Perfetto · Chrome

BINDINGS · Python / C++ 边界

_tracesmith.so
pybind11 native extension

CORE LIBRARY · C++17

Capture Module
CUPTIProfiler · MCPTIProfiler
Metal · Ascend · ROCm · eBPF

Common + Format
RingBuffer SPSC · StackCapture
SBTWriter / SBTReader

State Module
GPUStateMachine
TimelineBuilder · PerfettoExporter

Replay Module
ReplayEngine · StreamScheduler
DeterminismChecker · FrameCapture

CROSS-CUTTING · 集群与调试

replay / debug feedback

Cluster Module
GPUTopology · TimeSync
NCCLTracker

GDB / Tracy
RSPHandler · GPUDebugEngine
TracyExporter

GPU BACKENDS · 系统接口

NVIDIA
CUDA · CUPTI · nsys

MetaX
MACA · MCPTI · mcTracer

Apple / AMD
Metal · xctrace · ROCm

Huawei / Linux
CANN · msprof · eBPF

OUTPUTS · Trace 产物

trace.sbt
compact binary

trace.json / .perfetto
Perfetto timeline

graph.dot / ASCII
analysis views

主链：GPU event → lock-free ring → SBT/state → replay 或 Perfetto；Cluster / GDB 为旁路能力。